Veer Narmad South Gujarat University, Surat

Computer Science and Information Technology Faculty

M.Sc. (Computer Application)(Honours)

902 Robotic Process Automation (RPA)

UNIT2

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Syllabus of UNIT 2

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Q.1 Give overview of UiPath and UiPath orchesstrator. ?

Answer :

Robotic Process Automation (RPA):
UiPath is a software platform that enables businesses to automate repetitive tasks using robots. This helps improve efficiency and accuracy.

UserFriendly Interface: It provides a draganddrop interface, making it easy for users without extensive programming knowledge to create automation workflows.
Versatile Automation: UiPath can automate tasks across various applications, including web browsers, desktop applications, and even mainframe systems.

Key Components of UiPath
UiPath Studio: This is where you design and develop automation workflows. You can create processes by visually laying out steps.

UiPath Robot: Once a workflow is created, it is run by the UiPath Robot. There are two types:
Attended Robots: These work alongside humans and assist in tasks.
Unattended Robots: These run independently without human intervention.

UiPath Orchestrator:
This is a web based application used to manage, monitor, and deploy robots and workflows.

Central Management Console: Orchestrator serves as the central hub for managing all automation processes. It allows users to deploy, schedule, and monitor robots from a single platform.
Robot Management: You can manage and configure multiple robots. This includes assigning tasks and ensuring robots are up and running.
Job Scheduling: Orchestrator enables you to schedule when specific tasks or workflows should run, optimizing resource use.
Monitoring and Logging: It provides detailed logs of robot activities, which helps in troubleshooting and understanding performance metrics.
Security and Permissions: You can set user roles and permissions to control who can access and manage automation processes.

Benefits of Using UiPath and Orchestrator :
Increased Efficiency: Automating repetitive tasks saves time, allowing employees to focus on more strategic work.

Improved Accuracy: Robots perform tasks with high accuracy, reducing the risk of human error.
Scalability: As business needs grow, you can easily add more robots and workflows.
Cost Effective: Reducing manual work can lead to significant cost savings over time.
Enhanced Compliance: Automated processes can be designed to adhere strictly to regulations and standards, making compliance easier.

Use Cases for UiPath
Data Entry: Automating the input of data into systems.
Invoice Processing: Extracting data from invoices and entering it into accounting software.
Customer Service: Automating responses to common customer inquiries or processing requests.
Reporting: Gathering and compiling data for reports without manual intervention.

Conclusion:
UiPath and its Orchestrator provide powerful tools for businesses looking to embrace automation. By using these tools, companies can improve efficiency, reduce errors, and save costs, all while maintaining control and oversight of their automated processes.

Q.2 Explain recording and editing automation workflows of UiPath.
Answer :

Recording in UiPath:
Recording in UiPath is a feature that allows you to capture user interactions with applications to create automation workflows automatically. This is particularly useful for repetitive tasks that require several steps.

Types of Recording in UiPath
UiPath offers different types of recording options:

Basic Recording:
Ideal for simple tasks.
Captures mouse clicks and keyboard inputs in a straightforward manner.

Desktop Recording:
Designed for desktop applications.
Provides more detailed capturing, including full automation of desktop applications.

Web Recording:
Tailored for web applications.
Optimizes automation for actions like clicking buttons, entering data into forms, and scraping data from web pages.

Citrix Recording:
Used for virtualized environments, such as Citrix.
Captures actions in applications that are running on virtual desktops.

Steps for Recording Automation Workflows
Here's how to use the recording feature in UiPath:

Step 1: Open UiPath Studio
Launch UiPath Studio, the environment where you design automation workflows.

Step 2: Select the Recording Type
Click on the "Recording" option in the toolbar.
Choose the appropriate recording type based on the application (Basic, Desktop, Web, or Citrix).

Step 3: Start Recording
Click the "Start Recording" button.
Interact with the application as you normally would (clicking buttons, entering text, etc.).
UiPath will record these actions.

Step 4: Stop Recording
Once you have completed the task, stop the recording.
UiPath will generate a sequence of activities that correspond to the actions you performed.

Editing Automation Workflows
After recording, you may need to edit the workflow to refine it or add additional logic.

Step 1: Access the Workflow
After stopping the recording, you will see the generated workflow in UiPath Studio.
Open the workflow if it's not already visible.

Step 2: Review the Activities
Examine the activities generated by the recording. These will be displayed in a flowchart or sequence format.
Each action you took will be represented by an activity (e.g., Click, Type Into).

Step 3: Modify Activities
Edit Properties: Click on an activity to view its properties in the Properties pane. You can change settings like selector properties, timeouts, and input values.
Add New Activities: Use the Activities panel to drag and drop additional activities into your workflow. For example, you might want to add a decision (If activity) or loops (For Each).
Rearrange Activities: You can drag activities to change their order, ensuring the workflow follows the correct logic.

Step 4: Test the Workflow
Use the "Run" button to test your workflow and see if it behaves as expected.
Monitor for any errors or issues, and make adjustments as necessary.

Step 5: Save and Publish
Once satisfied with your workflow, save your project.
If you're ready to deploy it, you can publish the workflow to UiPath Orchestrator for broader use.

Best Practices for Recording and Editing
Plan Ahead: Before recording, have a clear understanding of the task to optimize your workflow design.
Minimize Unnecessary Actions: Try to perform only the necessary steps during recording to avoid clutter in the workflow.
Comment Your Workflow: Add comments in UiPath Studio to explain complex sections or decisions in your automation.
Use Selectors Wisely: Ensure that selectors (which identify UI elements) are reliable and specific to avoid errors during execution.

Conclusion
Recording and editing workflows in UiPath makes it easier to automate tasks efficiently. The recording feature allows you to quickly create workflows, while editing gives you the flexibility to refine them. With practice, you'll become adept at creating effective automations that enhance productivity.

Q.3 Describe debugging and troubleshooting automation workflow.

Answer :

1. Importance of Debugging and Troubleshooting
Error Prevention: Identifying issues before deploying can prevent failures in production.
Efficiency: Debugging helps optimize workflows, making them run faster and more reliably.
User Confidence: A welltested automation process builds trust with users and stakeholders.

2. Common Types of Errors
Syntax Errors: Mistakes in the programming structure that prevent the workflow from running.
Logical Errors: The workflow runs but produces incorrect results due to flaws in logic.
Runtime Errors: Issues that occur while the workflow is executing, such as elements not found or timeouts.
Selector Issues: Problems with UI elements not being recognized due to changes in the application.

3. Debugging Tools in UiPath
UiPath provides several builtin tools to help with debugging:

Breakpoints: You can set breakpoints in your workflow, which allows you to pause execution at specific points to inspect variables and the workflow state.
Debug Mode: Running the workflow in Debug mode enables stepbystep execution. You can see how each activity is processed.
Watch Panel: This allows you to monitor the values of specific variables while debugging.
Output Panel: Displays logs and messages that can help diagnose issues.
Step Into, Step Over, and Step Out: These options let you control the flow of execution during debugging, allowing you to navigate through the activities more precisely.

4. Steps for Debugging a Workflow
Step 1: Set Breakpoints
Click on the margin next to an activity to set a breakpoint. This will halt execution at that point, allowing you to examine the state of your workflow.

Step 2: Run in Debug Mode
Click the “Debug” button in UiPath Studio to start executing the workflow in debug mode. The execution will pause at breakpoints.

Step 3: Inspect Variables
When execution pauses, use the Watch Panel to monitor variable values. Check if they hold the expected data at each breakpoint.

Step 4: Step Through Activities
Use Step Into to execute the current activity and move to the next one, or Step Over to skip the current activity.
This helps you observe how the workflow behaves with each action.

Step 5: Analyze Logs and Output
Check the Output Panel for messages and logs. This can provide insights into where errors are occurring.

5. Troubleshooting Common Issues
Selector Issues
Review Selectors: Ensure that the selectors are correctly identifying UI elements. Use the UiExplorer tool to validate and refine selectors.
Use Wildcards: If elements have dynamic attributes, consider using wildcards to improve selector robustness.

Handling Exceptions
TryCatch Activities: Use TryCatch blocks to handle exceptions gracefully. You can define alternative actions if an error occurs.
Log Exceptions: Within the Catch section, log the exception details to help diagnose the problem later.

Workflow Logic
Review Logic Flow: Ensure that the sequence of activities makes sense. Use flowcharts for visual representation, if necessary.
Test Conditions: For conditional activities (like If statements), ensure that the conditions are set correctly and are being evaluated as expected.

6. Best Practices for Debugging and Troubleshooting
Modular Design: Break down workflows into smaller, manageable components. This makes debugging easier.
Regular Testing: Continuously test workflows during development to catch errors early.
Documentation: Document your workflows, especially complex logic, to aid in troubleshooting later.
Collaborate: Work with peers to review your workflows. Fresh eyes can catch issues you might overlook.

Conclusion:
Debugging and troubleshooting are essential skills when working with UiPath automation workflows. By effectively using debugging tools and following structured approaches, you can identify and resolve issues quickly, ensuring that your automations run reliably and efficiently. With practice, you'll become more proficient at diagnosing and fixing problems, leading to more robust automation solutions.

Q.4 Describe Ribbon and Universal search Box.
Answer :

1. Ribbon in UiPath

The Ribbon is a user interface element in UiPath Studio that organizes various tools and features. It is located at the top of the UiPath Studio window and consists of multiple tabs, each containing relevant commands and options.

Key Features of the Ribbon

Tabs: The Ribbon is divided into several tabs, each dedicated to a specific area of functionality. Common tabs include:
Home: Contains general tools for creating and managing projects, such as new file creation and opening existing files.

Design: Offers tools and activities for building workflows, including the ability to add different types of activities, sequences, and flowcharts.
Debug: Provides options for debugging your workflows, including setting breakpoints, running in debug mode, and stepping through activities.
Publish: Allows you to publish your automation projects to UiPath Orchestrator or other destinations.

Groups: Within each tab, commands are organized into groups. For example, in the Design tab, you might find groups for Control Flow, Data Manipulation, and User Interface activities.

Commands: Each command (button, dropdown, etc.) allows you to perform specific actions, such as adding activities, saving your project, or running the workflow.

Benefits of the Ribbon
UserFriendly: The Ribbon interface is designed to be intuitive, making it easier for users to find and use features.
Quick Access: With all essential tools consolidated in one area, users can quickly navigate and access what they need without digging through menus.
Contextual Functions: Depending on the selected tab, the commands presented are relevant to the tasks being performed.

2. Universal Search Box

The Universal Search Box is a powerful feature in UiPath Studio that enhances productivity by allowing users to quickly find activities, variables, and other elements within their projects.

Key Features of the Universal Search Box

Quick Access: Located at the top right corner of UiPath Studio, the search box enables users to type keywords to find specific activities, workflows, or items in the project.

Search Types: The Universal Search Box can be used to search for:
Activities: Type the name of an activity to find it quickly, even if it's buried in a long list.
Variables: Find specific variables defined within the project without scrolling through numerous panels.
Workflows and Files: Search for specific workflows, XAML files, or other components in the project.

Dynamic Suggestions: As you type, the search box provides dynamic suggestions based on the entered text. This helps you quickly locate the right item without needing to know the exact name.

Benefits of the Universal Search Box:

Increased Efficiency: Users can spend less time navigating through the interface and more time focused on building their automations.
Enhanced Productivity: Quickly finding items means you can work more fluidly and reduce disruptions during the development process.
Intuitive Interface: The search functionality feels natural, as many users are accustomed to using search boxes in other applications.

Conclusion:

Both the Ribbon and the Universal Search Box in UiPath Studio are designed to enhance user experience and streamline the development process. The Ribbon organizes essential tools in a logical, accessible manner, while the Universal Search Box provides a quick way to find activities and components. Together, they help users efficiently build, debug, and manage their automation projects.

Q.5 Explain Activities Panel, Design Panel and Library Panel of UiPath.

Answer :

1. Activities Panel

The Activities Panel is a core component of UiPath Studio, where you can find all the prebuilt activities available for building automation workflows.

Key Features:
Categories: Activities are organized into categories such as UI Automation, System, Programming, Data Manipulation, and more. This makes it easier to locate specific activities based on the type of automation you are working on.
Search Functionality: You can quickly find an activity by typing its name in the search bar within the Activities Panel.
DragandDrop Interface: You can simply drag an activity from the panel into your workflow design area (Design Panel) to incorporate it into your automation process.
Tooltips and Descriptions: Hovering over an activity provides a brief description, helping you understand what each activity does without needing to look it up separately.

Benefits:
Ease of Use: The structured organization allows users to easily navigate and locate activities.
Rapid Development: Dragging activities into workflows speeds up the process of building automation.
Comprehensive Options: A wide variety of activities cater to numerous automation needs, from simple data entry to complex business logic.

2. Design Panel

The Design Panel is where you create and visualize your automation workflows. It acts as the main workspace in UiPath Studio.

Key Features:

Workflow Area: This is the central area where you design your workflows, which can be in sequence or flowchart format. You can visually represent the steps in your automation process.

Adding Activities: After dragging activities from the Activities Panel, you can arrange them in the order of execution, connecting them as needed.

Properties Pane: When you select an activity, the Properties Pane appears on the right, allowing you to configure settings and parameters for that specific activity.

Annotations and Comments: You can add annotations and comments within the workflow to clarify complex logic or provide context, which helps in maintaining the workflow.

Benefits:

Visual Representation: The ability to see the workflow visually aids in understanding and designing complex automation processes.

Organized Structure: Users can easily follow the flow of the process, making it simpler to debug and modify workflows.

Customizable: The design area allows for flexibility in arranging activities, providing control over the structure of the workflow.

3. Library Panel

The Library Panel allows you to manage and create reusable components (libraries) in UiPath.

Key Features:

Reusability: You can create reusable workflows and components that can be used across multiple projects, promoting consistency and saving development time.

Custom Activities: Libraries can contain custom activities that you design, which can be shared with other users or teams.

Importing Libraries: You can import existing libraries from other projects or use libraries created by the UiPath community.

Version Control: You can manage versions of your libraries, allowing for easy updates and maintenance.

Benefits:

Efficiency: By reusing components, you avoid redundant work, making the development process faster and more efficient.

Collaboration: Shared libraries enable teams to work together effectively, maintaining uniformity in automation solutions.

Quality Control: Using tested and verified libraries can help reduce errors in your workflows.

Conclusion

The Activities Panel, Design Panel, and Library Panel in UiPath Studio are integral parts of the automation development environment. The Activities Panel provides access to a wide range of activities, the Design Panel allows you to build and visualize workflows, and the Library Panel promotes reusability and collaboration. Together, these panels create a streamlined and efficient workflow automation experience.

Q.6 Explain Properties Panel and Outline Panel of UiPath.
Answer :

1. Properties Panel

The Properties Panel in UiPath Studio provides detailed settings and configuration options for the selected activity within your workflow.

Key Features:
Contextual Settings: The Properties Panel displays different options depending on which activity you have selected. Each activity has its own set of properties that can be configured.
Common Properties: Some common properties you might see include:
Name: The name of the activity, which you can customize.
Input/Output Parameters: Fields where you can define inputs needed for the activity to execute and the outputs it will generate.
Timeout: Specifies how long to wait for the activity to complete before throwing an error.
Selectors: For UI activities, this is where you specify which UI element the activity should interact with.
Variables and Arguments: You can link variables and arguments to activity properties, allowing dynamic input and output handling.

Benefits:
Detailed Configuration: Users can finetune activities to meet specific requirements, enhancing the automation's effectiveness.
Ease of Use: The organized layout of the Properties Panel simplifies the task of setting up complex activities.
Error Reduction: By providing clear fields for input, the Properties Panel helps reduce errors related to incorrect configurations.

2. Outline Panel

The Outline Panel offers a structured view of the components within your workflow, helping you manage and navigate through them easily.

Key Features:
Hierarchical Structure: It displays a hierarchical representation of all the activities and elements in your workflow. This includes sequences, flowcharts, and individual activities.
Navigation: You can click on any item in the Outline Panel to quickly jump to that part of the workflow in the Design Panel.
Grouping: Activities can be grouped or nested, allowing for better organization of complex workflows. The Outline Panel helps visualize these groupings.
Search Functionality: The Outline Panel may also include a search feature to quickly locate specific activities or components.

Benefits:
Simplified Navigation: The Outline Panel makes it easy to find and access different parts of your workflow, especially in larger projects.
Organizational Clarity: By providing a clear structure, it helps users maintain organized workflows and understand the overall flow of the process.
Quick Edits: You can select activities directly from the Outline Panel to edit or configure them without needing to scroll through the Design Panel.

Conclusion

The Properties Panel and Outline Panel in UiPath are essential tools that enhance the workflow development experience. The Properties Panel allows for detailed configuration of each activity, ensuring that users can customize their automations effectively. Meanwhile, the Outline Panel provides a clear, organized view of the workflow structure, making it easier to navigate and manage complex automation projects.

Q.7 Describe features of Output Panel and Control panel of UiPath.
Answer :

1. Output Panel

The Output Panel in UiPath Studio is a crucial component for monitoring and debugging your automation workflows. It displays messages and logs generated during the execution of your workflows.

Key Features:
Execution Logs: The Output Panel shows logs related to the execution of your workflow, including information about each activity as it runs. This can include status messages such as "Started," "Completed," or "Failed."
Error Messages: If an error occurs during execution, the Output Panel will display error messages and stack traces, which help in identifying what went wrong and where.

Debugging Information: When running a workflow in Debug mode, the Output Panel can provide additional debugging information, including variable values and execution paths.
Custom Messages: You can use the Write Line or Log Message activities to output custom messages to the Output Panel. This is useful for tracking the workflow's progress or debugging.
Filtering Options: The Output Panel may have filtering options that allow you to view only specific types of messages (e.g., errors, warnings, or informational messages).

Benefits:
RealTime Monitoring: The Output Panel enables you to monitor the progress and status of your workflow in realtime.
Effective Debugging: With error and execution logs readily available, debugging becomes easier, allowing for quick identification and resolution of issues.
Enhanced Clarity: Custom messages provide clarity on workflow operations, making it easier to understand how the automation is functioning.

2. Control Panel

The Control Panel (often referred to as the Execution Control Panel) is a part of UiPath Studio that allows you to control the execution of your workflows. It includes buttons and controls for managing workflow execution.

Key Features:
Run Button: This allows you to execute the entire workflow in either normal mode or Debug mode.
Pause and Stop: You can pause or stop the execution of a running workflow, providing control over longrunning processes.
Step Into, Step Over, and Step Out: These options are used during debugging:
Step Into: Executes the current activity and moves into any nested activities.
Step Over: Executes the current activity and moves to the next one without entering nested activities.
Step Out: Completes the execution of the current activity and returns to the parent activity.
Reset: This option allows you to reset the execution state of the workflow, clearing any stored data or variables.

Benefits:
User Control: The Control Panel gives users the ability to manage workflow execution easily, enhancing the debugging and testing process.
Efficiency: Quick access to control options speeds up the development and testing cycle, allowing for iterative testing and adjustments.
Clarity During Debugging: The step control features help users understand the flow of execution and inspect individual activities closely.

Conclusion

The Output Panel and Control Panel in UiPath Studio are essential for monitoring, debugging, and managing workflow execution. The Output Panel provides realtime logs and error messages, aiding in troubleshooting, while the Control Panel allows users to execute, pause, and step through workflows efficiently. Together, these panels enhance the overall development experience in UiPath.

M.C.Q.
UNIT – 2

1. What is the primary function of UiPath Studio?
A) Manage automation robots
B) Design and develop automation workflows
C) Monitor workflow execution
D) Generate reports
Correct Answer: B

2. What does UiPath Orchestrator primarily manage?
A) Data extraction
B) Workflow design
C) Robot deployment and scheduling
D) User interface
Correct Answer: C

3. Which of the following is NOT a component of UiPath Studio?
A) Activities Panel
B) Properties Panel
C) Robot Manager
D) Design Panel
Correct Answer: C

4. Which version of .NET is required to run UiPath Studio?
A) .NET Framework 4.5
B) .NET Framework 4.6.1
C) .NET Framework 3.5
D) .NET Core
Correct Answer: B

5. What is the first step in installing UiPath Studio?
A) Configure Orchestrator
B) Download the installer from the UiPath website
C) Set up a robot
D) Create a new project
Correct Answer: B

6. Which panel in UiPath Studio allows users to search for specific activities?
A) Properties Panel
B) Output Panel
C) Activities Panel
D) Design Panel
Correct Answer: C

7. What is the purpose of the Ribbon in UiPath Studio?
A) Displays log messages
B) Organizes tools and commands
C) Manages robot schedules
D) Provides a coding environment
Correct Answer: B

8. What type of recording captures user interactions with web applications?
A) Basic Recording
B) Desktop Recording
C) Web Recording
D) Citrix Recording
Correct Answer: C

9. After recording a workflow, what can you do to refine it?
A) Publish it immediately
B) Edit activities in the Design Panel
C) Delete all activities
D) Set it to run unattended
Correct Answer: B

10. Which activity is commonly used to input data into a text field?
A) Click
B) Type Into
C) Assign
D) Log Message
Correct Answer: B

11. What is the purpose of variables in UiPath?
A) Store user interface elements

B) Hold data values during automation
C) Control workflow execution
D) Generate reports
Correct Answer: B

12. What does the Try Catch activity do in UiPath?
A) Terminates the workflow
B) Attempts to execute a block of code and handles exceptions
C) Logs all activities
D) Runs the workflow in debug mode
Correct Answer: B

13. What panel provides realtime logs of the workflow execution?
A) Properties Panel
B) Output Panel
C) Activities Panel
D) Control Panel
Correct Answer: B

14. Which tab on the Ribbon would you use to publish your workflow?
A) Home
B) Design
C) Debug
D) Publish
Correct Answer: D

15. What can you find in the Debug tab of the Ribbon?
A) Workflow settings
B) Execution control options
C) Activity properties
D) Activity library
Correct Answer: B

16. What is the primary use of the Universal Search Box in UiPath Studio?
A) Search for external libraries
B) Find specific activities or components
C) Run the workflow
D) Access project settings
Correct Answer: B

17. Can the Universal Search Box help in locating variables?
A) Yes
B) No
C) Only if they are global

D) Only if they are in the Library Panel
Correct Answer: A

18. Which panel is used to organize and access reusable components?
A) Activities Panel
B) Library Panel
C) Properties Panel
D) Output Panel
Correct Answer: B

19. What does the Project Panel display?
A) All activities available in UiPath
B) The hierarchy of files and resources in the current project
C) Workflow execution logs
D) Active robots
Correct Answer: B

20. What information is primarily shown in the Properties Panel?
A) Execution logs
B) Settings and configurations for the selected activity
C) Available activities
D) Workflow hierarchy
Correct Answer: B

21. What does the Outline Panel help you manage?
A) Activity settings
B) Workflow structure and navigation
C) Robot schedules
D) Installed libraries
Correct Answer: B

22. What does the Control Panel allow you to do during debugging?
A) Set breakpoints
B) Control the execution of the workflow
C) Search for activities
D) View execution logs
Correct Answer: B

23. Which activity would you use to scrape data from a website?
A) Data Scraping
B) Read Range
C) Click
D) Type Into
Correct Answer: A

24. What type of robot executes tasks without human intervention?
A) Attended Robot
B) Unattended Robot
C) Assistant Robot
D) Hybrid Robot
Correct Answer: B

25. Which panel would you primarily use to create complex workflows?
A) Activities Panel
B) Design Panel
C) Properties Panel
D) Output Panel
Correct Answer: B

26. What is the function of the Write Line activity?
A) To log information to the Output Panel
B) To write data to an external file
C) To display a message box
D) To input data into a form
Correct Answer: A

27. Which of the following is a key benefit of using UiPath Orchestrator?
A) Designing workflows
B) Monitoring robot performance and scheduling tasks
C) Recording user interactions
D) Accessing the Activities Panel
Correct Answer: B

28. What does a flowchart in UiPath represent?
A) A linear sequence of activities
B) Conditional branching and loops
C) Only data manipulations
D) A list of activities
Correct Answer: B

29. Which activity is used to click a UI element?
A) Type Into
B) Click
C) Select Item
D) Get Text
Correct Answer: B

30. What is the purpose of the Selector in UiPath activities?

A) Define the workflow structure
B) Identify UI elements for automation
C) Store variable values
D) Manage project settings
Correct Answer: B